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Original research article

High-performance dual-channel PCF-SPR sensor for RI interrogation and numerical analysis of cancerous cells, urinary tract infection pathogens, and dengue-affected blood components

Adib Hassan [a,*], Md. Tanjim Hasan [a], Shaharior Anik [b], Tanvir Ahmed [c], Alok Kumar Paul [c]

[a] Department of Electrical & Electronic Engineering (EEE), Rajshahi University of Engineering & Technology (RUET), Rajshahi 6204, Bangladesh
[b] Department of Electrical & Electronic Engineering, Green University of Bangladesh, Dhaka 1216, Bangladesh
[c] Department of Electrical & Electronic Engineering (EEE) at Rajshahi University of Engineering & Technology (RUET), Rajshahi 6204, Bangladesh

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ABSTRACT

Label-free biosensing has made photonic crystal fiber (PCF)-based surface plasmon resonance (SPR) sensors increasingly popular. Yet many previously reported designs focused on detecting a single analyte within a specific refractive index (RI) range. Addressing this gap, we present a numerical investigation of a dual-core, dual-channel PCF-SPR sensor operating over a RI range of 1.17–1.41. The dual-channel arrangement enables simultaneous two-analyte sensing within a single fiber, reducing detection time and system complexity. Simulations performed with the finite element method (FEM) in COMSOL Multiphysics indicate a maximum wavelength sensitivity (WS) of 122,000 nm/RIU, an amplitude sensitivity (AS) of 832.18 RIU^{-1} , a figure of merit (FOM) of 739.39 RIU^{-1} , and a numerical resolution of 8.2×10^{-7} RIU. The sensor can also numerically discriminate six cancer cell types, three UTI pathogens, and dengue-related blood components, demonstrating the potential of the proposed architecture for comparative biomedical RI sensing.

1. Introduction

Photonics crystal fiber (PCF)-based surface plasmon resonance (SPR) sensing devices have attracted significant research attention owing to their unique ability to interrogate the refractive index (RI) of the sensing medium over a broad range and their highly tunable optical guiding characteristics [1,2]. Unlike typical optical fibers, PCFs have excellent light-guiding properties. Their relatively low fabrication cost and design flexibility make them highly attractive for diverse sensing applications. Also, PCF-based sensors offer maximum sensitivity, durability, and greater adaptability, making them a feasible solution for addressing industrial problems and delivering desired outcomes across diverse environments. PCF-based sensors provide a wide range of sensing applications, including medical diagnosis, chemical sensing, time-domain spectroscopy, water and environmental quality monitoring, food safety, biochemical and gaseous molecule identification, and so on [3,4]. With the advancement of modern technology, expectations for an improved and healthy quality of life have increased. Unfortunately, diseases like Cancer [5], Urinary Tract Infection (UTI),

* Corresponding author.
E-mail addresses: adibhassan014@gmail.com (A. Hassan), tjhasanonline@gmail.com (Md.T. Hasan), anik@eee.green.edu.bd (S. Anik), tanvir.ahmed@eee.ruet.ac.bd (T. Ahmed), alok@eee.ruet.ac.bd (A.K. Paul).

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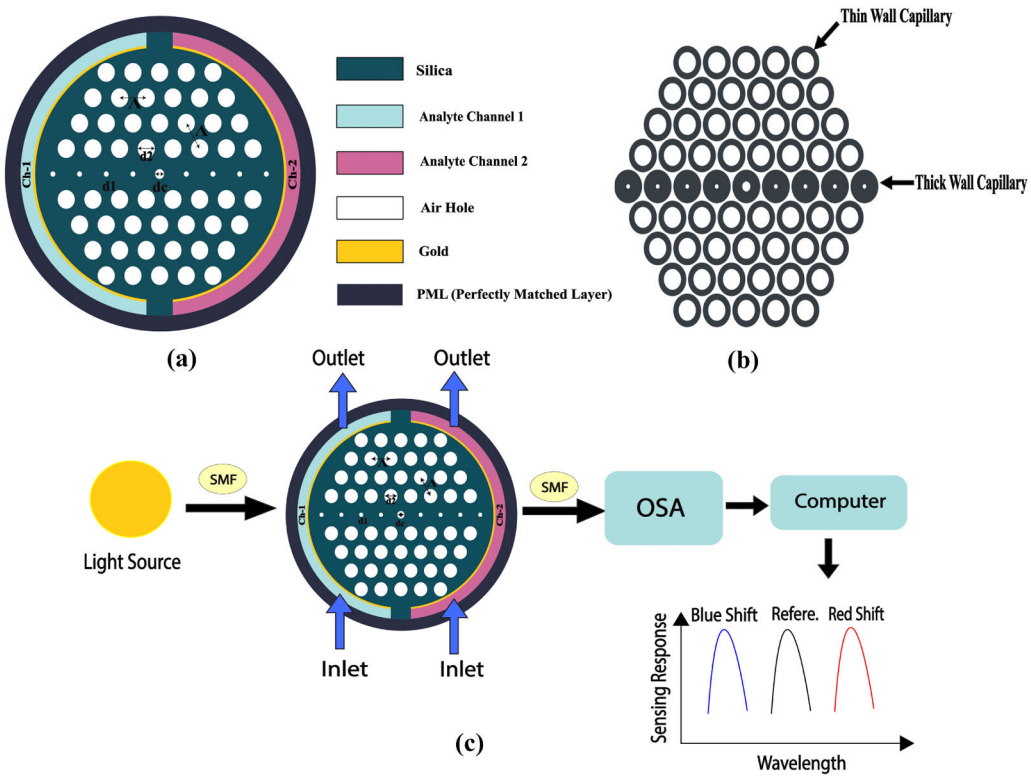


Fig. 1. (a) Schematic representation of the proposed dual-channel-based model; (b) Potential fiber configuration for fabrication using the stack and draw method; (c) Conceptual Sensing arrangement.

Tuberculosis [6], and dengue [7], are affecting people worldwide. Identification of these conditions early in a timely and cost-effective way can greatly improve survival and recovery. This is why, in recent years, a great deal of research has been directed toward designing highly efficient biosensors [8]. Bepare et al. in [9] suggested a single-core biosensor that can detect different cancerous cells. This biosensor demonstrated maximum wavelength sensitivity (WS) of 8571.42 nm/RIU for the MCF-7 and MDA-MB-231 cancer cells. Siddik et al. [7] modeled a SPR biosensor coated with gold for dengue virus detection. This sensor demonstrated 4857.14 nm/RIU WS for hemoglobin. Among the various sensing technologies developed, SPR has consistently stood out for its remarkable sensitivity, real-time response, and label-free sensing capability. SPR is produced by the collective free-electron oscillations at the metal dielectric junction. Direct coupling of incident light cannot excite SPR directly because its propagation constant does not match that of the SPR wave [10]. Selection of the correct plasmonic materials is essential for the development of practical sensors. When an electromagnetic field interacts with highly energized particles in plasmonic materials such as copper, aluminum, silver, and gold, it can give rise to surface plasmon-polaritons (SPPs) at interfaces between metals and dielectrics. This process converts a significant portion of the optical energy from the core into SPP mode, resulting in a sharper resonance peak [11]. Among these metals, silver is often considered better due to the absence of an interband transition and a sharper SPP resonance peak. However, silver can't be widely used in an aqueous environment due to rapid oxidation. Gold, in comparison, is chemically stable and represents a substantial shift in the loss peak as opposed to other materials [12].

A sensor with the capability of detecting multiple analytes can be classified as a multi-analyte sensor. Most previously published SPR sensors are limited to detecting a single analyte, while only a few designs support simultaneous detection of multiple analytes. Generally, multiple analyte detection is performed employing multiple channels. These designs provide several benefits, including detection of multiple samples on the same structure, reduced detection time, and lower overall costs by integrating several sensing elements on the same platform [13–15]. In previous years, several articles on PCF-based multi-channel sensors have been published. H. Md Abu et al. [16] proposed a dual-channel PCF-SPR sensor that exhibited a maximum amplitude sensitivity (AS) of 2338 RIU^{-1} and a WS of 32,000 nm/RIU. However, their structure is operable only within a narrow RI window of 1.33–1.45. Kawser et al. [12] reported a four-core multi-analyte sensor with a WS of 10,000 nm/RIU and an AS of 1204 RIU^{-1}. Divya et al. [6] designed a dual-channel SPR sensor with a WS of 10,000 nm/RIU and 8235.29 nm/RIU for channels 1 and 2. However, the internal multilayer plasmonic coating of this structure during the deposition process requires precise control. Furthermore, the Ag-based plasmonic layer is susceptible to oxidation and degradation. Kaur et al. in [17] demonstrated a two-channel PCF structure using an internal gold-coated channel, showcasing a WS of 1000 nm/RIU for ch-1 and 3750 nm/RIU for ch-2, respectively. However, the reported confinement loss of 5449 dB/cm is too high for practical fiber lengths.

This work presents a highly sensitive dual-core, dual-channel PCF-SPR architecture that numerically operates over a RI range of

1.17–1.41. Additionally, the numerical response of the sensor is further evaluated within the practical biosensing range of 1.335–1.401 for RI-based discrimination of normal and pathological samples associated with six cancer cell lines, three UTI pathogens, and dengue related blood components. Numerical analysis is performed utilizing finite element method (FEM) implemented in COMSOL Multiphysics v6.2, with a circular perfectly matched layer (PML) placed at the outer boundary. A thin gold film is coated on the outer cladding surface to realize the SPR effect. As the plasmonic layer is exposed to the surrounding medium, the proposed sensor enables direct interaction with the analyte through an external sensing mechanism.

2. Geometric design and numerical investigation

The geometry of the proposed dual-core, dual-channel fiber is shown in Fig. 1. Fig. 1(a) depicts a 2D cross-sectional view of the designed model. Fig. 1(b) presents a stacked layout of the fiber using thin and thick capillaries. In this analysis, we have used various materials, including silica, gold, and airholes. Circular airhole layers are organized in a hexagonal lattice arrangement to establish the cladding. Three distinct sizes of air holes, d_c, d_1, and d_2, are used for this model. Adjacent air holes in the structure maintain a uniform center-to-center spacing referred to as the pitch (Λ), which is 2.3 μm. An air hole with d_c= 0.7μm diameter is placed at the center of the core region. Diameter of the larger air hole structure is d_2 = 1.4 μm, and the smallest air hole diameter is d_1 = 0.35 μm. A circular perfectly matched layer (PML) with a thickness of 2.013 μm, equal to 10% of the PCF diameter of 20.13 μm, is employed at the outer surface to absorb radiated power. Fig. 1(c) presents the sensing arrangement employing the proposed design. Source light enters a standard single-mode fiber (SMF) and is delivered to the designed sensor through a splicing process, which builds up an evanescent field that drives plasmon resonance along the metal surface while the analyte enters through the inlet. Analyte A and Analyte B are introduced through two different inlets, and leave through their respective different Outlets; the two channels are separated by the silica bridges and do not mix. An optical spectrum analyzer (OSA) records the resulting spectrum, which is processed by a computer to reveal whether the resonance experiences a blue shift or red shift relative to the reference wavelength.

For fiber structure, fused silica (SiO_2) is used as the base material. Using Sellmeier equation, the RI of fused silica can be determined [18]:

$$n^2(\lambda) = 1 + \frac{E_1\lambda^2}{\lambda^2 - G_1} + \frac{E_2\lambda^2}{\lambda^2 - G_2} + \frac{E_3\lambda^2}{\lambda^2 - \quad G_3}$$

Here, the operating wavelength λ is expressed in μm. The Sellmeier coefficients are E_1 = 0.69616300, E_2 = 0.407942600, E_3 = 0.897479400, G_1 = 4.67914826 × 10^{-3} μm^2, G_2= 1.35120631 × 10^{-2} μm^2, and G_3= 97.9340025 μm^2.

The chemical vapor deposition (CVD) technique is used to coat the plasmonic material gold on the outer fiber surface. A t_g= 40 nm optimized gold layer is introduced here. The permittivity of the gold layer can be evaluated using the Drude–Lorentz model [19]:

$$\varepsilon_{Au} = \varepsilon_\infty - \frac{\omega_D^2}{\omega(\omega + j\gamma_D)} - \frac{\Delta\varepsilon\Omega_L^2}{(\omega^2 - \Omega_L^2) + j\Gamma_L\omega}$$

where ε_{Au} represents gold's permittivity, ε_∞ = 5.9673 represents permittivity at a higher frequency. The angular frequency is ω = 2πc/λ. The parameters ω_D and γ_D denote the plasma frequency and damping frequency, with respective values of $\omega_D/2\pi$ = 2113.6 THz and $\gamma_D/2\pi$= 15.92 THz. $\Delta\varepsilon$ = 1.09 represents the weighting factor. The spectral width is $\Gamma_L/2\pi$ = 104.86 THz and frequency of the Lorentz oscillator is $\Omega_L/2\pi$ = 650.07 THz.

The proposed structure design yields a dual-core arrangement. Initial optimization is performed at 0.65 μm operating wavelength with the RI of the analyte set at 1.35. To ensure numerical accuracy, a finer mesh was adopted following a mesh refinement test.

The performance of the SPR sensor can be analyzed by evaluating the loss of optical fiber, particularly the confinement loss (CL). The confinement loss of the core-guided mode is found by [20]:

$$CL(\mathrm{dB/cm}) = 8.686 \times \frac{2\pi}{\lambda} Im(n_{eff}) \quad \times 10^4$$

In this context, λ represents functional wavelength and Im(n_{eff}) is the imaginary component of the core mode's effective RI.

The resonance wavelength is defined as the wavelength of the CL maximum. Resonance wavelength shifts are caused by variations in analyte RI, enabling unknown analytes to be identified using wavelength interrogation. Sensor's wavelength sensitivity (WS) is computed as [20]:

$$WS(nm/RIU) = \frac{\Delta\lambda_{peak}}{\Delta n_a}$$

In this context, $\Delta\lambda_{peak}$ and Δn_adenote the peak shift of the resonance wavelength and alteration of analyte RIs. Besides WS, amplitude sensitivity (AS) is another key parameter in this study. AS can be calculated employing the amplitude interrogation method, as expressed below [21]:

$$AS(RIU^{-1}) = -\frac{1}{\alpha(\lambda, n_a)} \frac{\partial\alpha(\lambda, n_a)}{\partial n_a}$$

$\alpha(\lambda, n_a)$ represents the CL as a function of wavelength (λ) and RI of analyte is n_a, while $\partial\alpha(\lambda, n_a)$ signifies deviation in loss due to ∂n_a variation of analyte RI. Sensor performance can also be assessed by sensor resolution using [22].

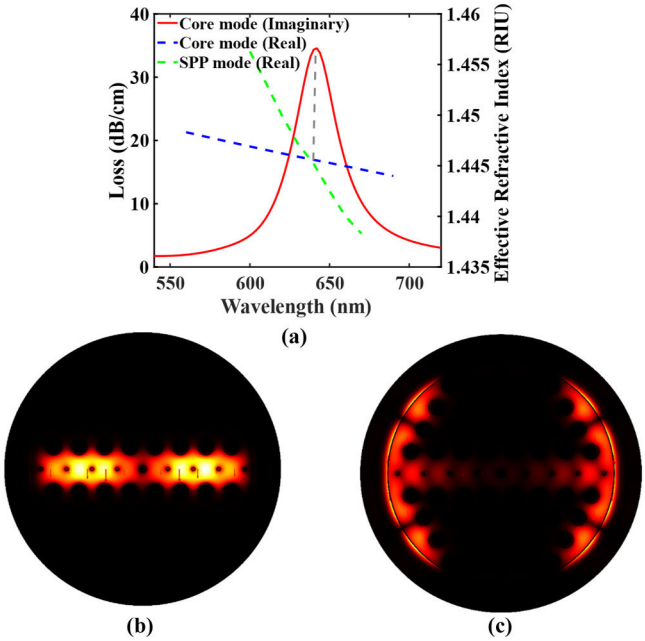


Fig. 2. (a) Phase-matching properties of the designed model for RI = 1.35, (b) Core mode, (c) SPP mode at 640 nm.

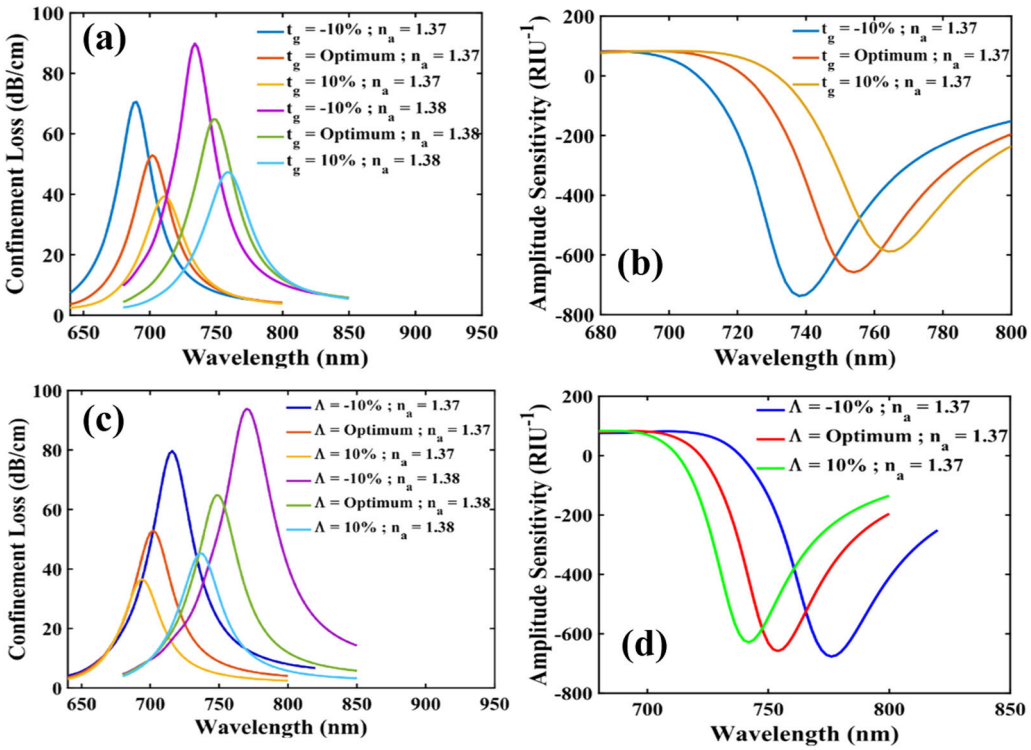


Fig. 3. (a), (b) Optimization of gold layer thickness, (c), (d) Pitch Optimization.

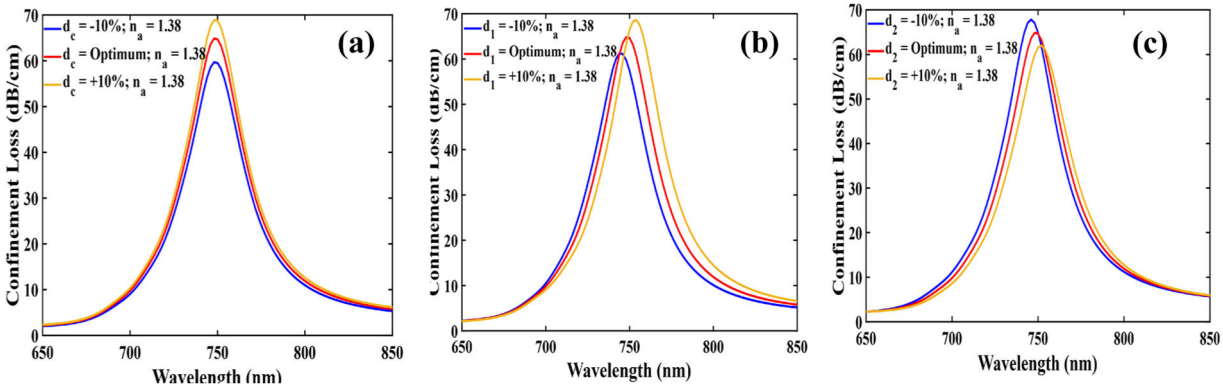


Fig. 4. Optimization of (a) d_c, (b) d_1, and (c) d_2 air hole diameters.

$$SR(RIU) = \frac{\Delta n_a \times \Delta\lambda_{\min}}{\Delta\lambda_{peak}}$$

The symbol $\Delta\lambda_{\min} = 0.1nm$ indicates the minimum detectable wavelength shift of the sensor. The figure of merit (FOM) is another significant sensor characteristic. This can be obtained using the subsequent equation [18]:

$$FOM = \frac{WS}{FWHM}$$

Here, FWHM denotes full width at half maximum calculated in nm.

3. General sensing performance of the proposed model

3.1. Electric field distribution and phase matching

Incident light propagating through the sensor core travels to the metal cladding part, stimulating free electrons and generating Surface Plasmon Waves (SPW) that give rise to a resonance peak [23]. The surrounding environment RI strongly influences the generation of SPWs and the variation in their sensitivity.

Phase-matching properties for y-polarized light between the core mode and the SPP mode are shown in Fig. 2(a), where both channels are at RI 1.35. The core mode (blue) and SPP mode (green) intersect at a wavelength of 640 nm. The RI of the core mode decreases steadily, whereas the RI of the SPP mode experiences an abrupt change. The resonance wavelength occurs at 640 nm, signifying maximum transfer of power from the core to the metal cladding, giving rise to the CL peak. Figs. 2(b) and 2(c) exhibit the electric field intensity distribution of the core and SPP mode for the proposed model. Fig. 2(b) indicates that the core mode is well confined within the guiding region. Whereas in Fig. 2(c) the SPP mode is confined at the metal-dielectric interface.

3.2. Structural parameters optimization

The resonance behaviour of the sensor is sensitive to the gold layer thickness (t_g) and the pitch (Λ), as both parameters alter the phase-matching condition and the strength of core-to-plasmon coupling. The gold layer has direct involvement in the generation of SPWs [24]; hence, the damping effect due to optimization determines the change in response in CL depth or wavelength shift. Similarly, pitch optimizing, as it is influential in the design of the core, which leads to energy transfer, leading to coupling. Gold layer thickness and pitch optimization are done while keeping the same RI at both channels.

Fig. 3(a) exhibits the ±10%, optimum t_g for RI 1.37 and 1.38. It demonstrates that as gold thickness increases, the CL depth has a loss due to weak coupling as the SPP effective mode index becomes more bound. It also shows red shift at the same RI as the thickness of gold increases. While Fig. 3(c) exhibits the ±10%, optimum pitch, Λ for RI 1.37 and 1.38. Here, as well, the CL depth reduction occurs due to pitch thickness, leading to the core mode effective index becoming more bound. In this case, there is a blue shift as pitch distance increases within the same RI. Figs. 3(b) and 3(d) exhibit the Amplitude sensitivity curve for RI 1.37 for gold layer optimization and pitch optimization, respectively. In both cases, the sharpness decays as the value of t_g or Λ increases.

Fig. 4 exhibits the variation of the three airhole diameters d_c, d_1, and d_2 for ±10% relative to their nominal values, at the RI of 1.38 while the other parameters remain fixed. As seen from Fig. 4(a), the CL peaks for the center horizontal air holes increase as the diameter d_c increases. Fig. 4(b) shows that with scaling the airholes of d_1 diameter, the CL peak increases, and also a slight red shift is observed. But, as seen in Fig. 4(c), the CL peaks decreases as the d_2 airhole diameter increases.

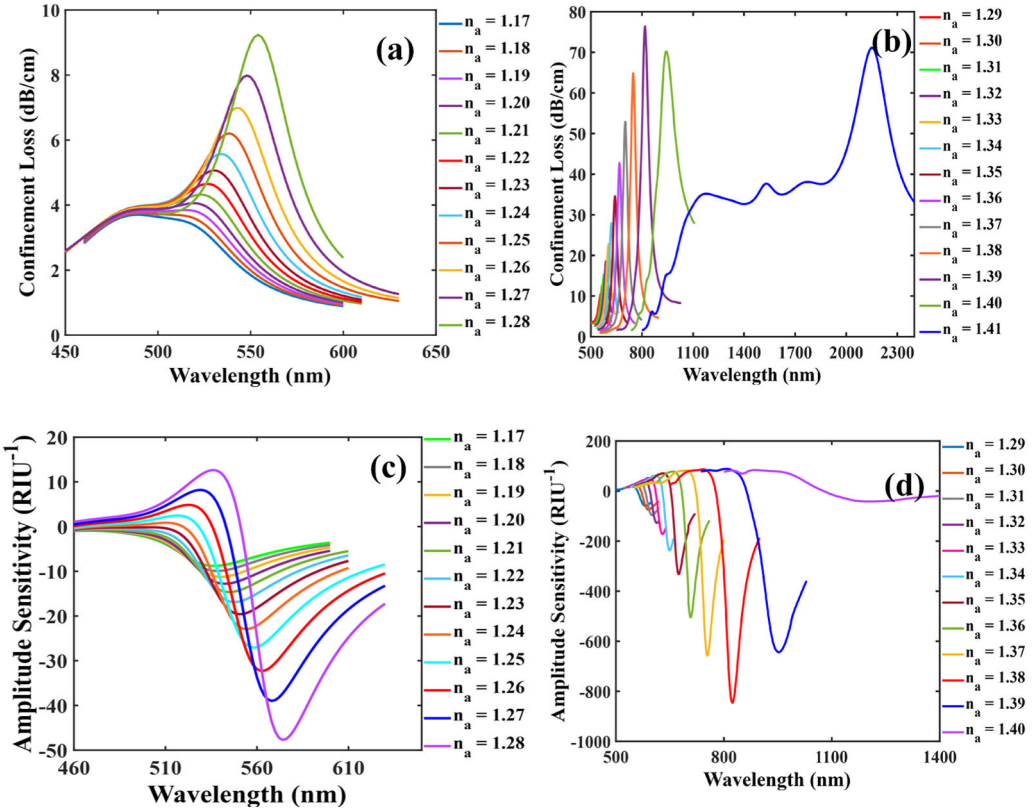


Fig. 5. CL curves for RI (a) 1.17–1.28 and (b) 1.29–1.41, AS curves for RI (c) 1.17–1.28 and (d) 1.29–1.40.

Table 1
Performance Analysis for Different Analyte Refractive Index.

RI	CL (dB/cm)	AS (RIU^{-1})	WS (nm/RIU)	FWHM (nm)	FOM (RIU^{-1})	SR (10^{-5})
1.17	3.70	8.02	200	100	2	50
1.18	3.76	9.90	200	102	2	50
1.19	3.85	11.23	600	106	5.66	16.7
1.20	4.06	12.71	400	108	3.70	25
1.21	4.32	14.30	200	108	2	50
1.22	4.65	16.78	400	108	3.70	25
1.23	5.06	19.58	400	107	3.73	25
1.24	5.57	22.33	400	108	3.70	25
1.25	6.21	26.96	400	106	3.77	25
1.26	6.99	31.87	600	99	6.06	16.7
1.27	7.99	38.79	600	64	9.37	16.7
1.28	8.98	45.89	1000	53	18.86	10
1.29	10.80	58.99	1000	48	20.83	10
1.30	12.79	75.03	1000	45	22.22	10
1.31	15.29	96.72	1000	42	23.80	10
1.32	18.64	117.74	1000	39	25.64	10
1.33	21.83	171.97	2000	36	55.55	5
1.34	27.99	236.64	2000	36	55.55	5
1.35	34.36	309.40	3000	35	85.71	3.33
1.36	42.09	463.95	3000	36	83.33	3.33
1.37	52.15	631.35	5000	37	135.14	2
1.38	64.67	832.18	7000	42	166.67	1.43
1.39	75.83	644.27	12000	57	210.52	0.83
1.40	70.11	42.13	122000	165	739.39	0.082
1.41	71.12	NA	NA	NA	NA	NA

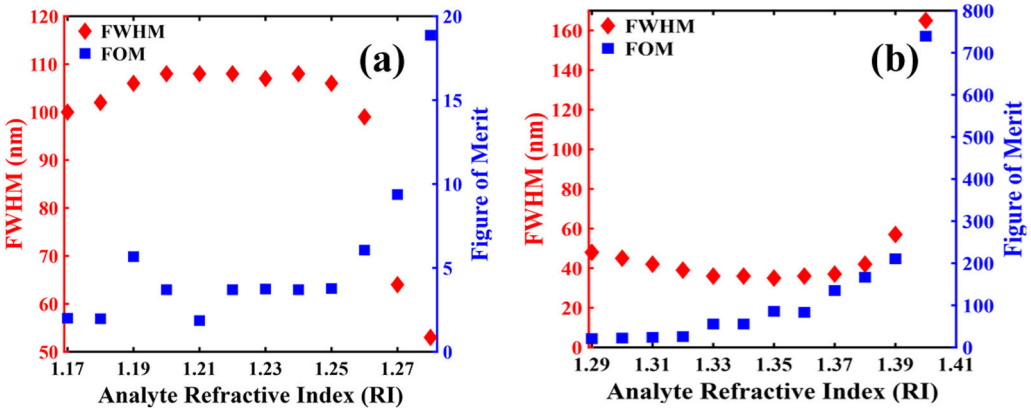


Fig. 6. FWHM and FOM of the proposed model for RI of different analytes (a) 1.17–1.28 and (b) 1.29–1.40.

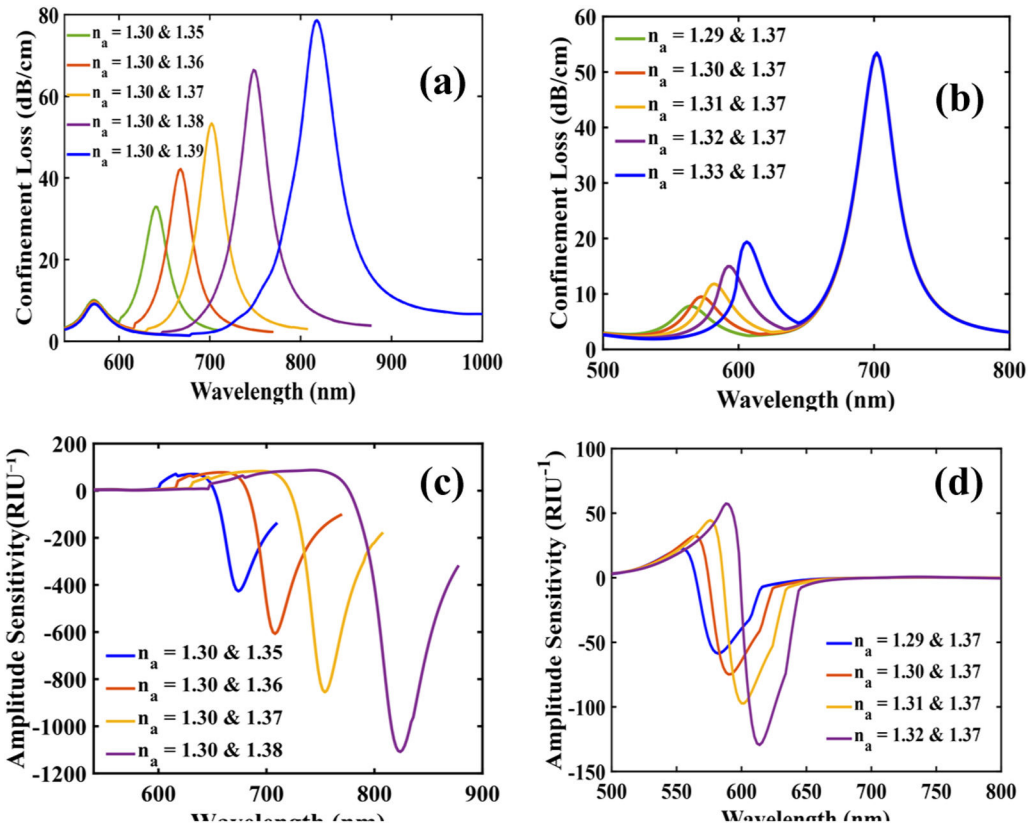


Fig. 7. (a) and (c) Characterization of CL curves and AS curves with Ch-1 fixed at one analyte RI and Ch-2 analyte RI varied, (b) and (d) Characterization of CL curves and AS curves with Ch-1 varied with analyte RI and Ch-2 analyte RI fixed.

3.3. RI variation by keeping the same RI in both channels

An inquiry into the performance of the sensor for analyte RI variation is conducted next. Same RI is kept on both channels, and y-polarized light for each RI varying from 1.17 to 1.41 with an increment of 0.01 RIU is demonstrated. CL is calculated by utilizing equation (3), and AS is calculated by utilizing equation (5). The loss peak and Amplitude Sensitivity graphs in the range of 1.17–1.28 are exhibited in Figs. 5(a) and 5(c). The CL loss peak for this range starts from 3.7 dB/cm up to 8.98 dB/cm for 1.17 RI to 1.28 RI chronologically. Whereas for the AS graph in Fig. 5(c), the range starts from 8.02 RIU^{-1} to 45.89 RIU^{-1} for the same range. Wavelength range for this set is from 450 nm to 650 nm. Figs. 5(b) and 5(d) exhibit the CL loss peak and AS graphs for the range of 1.29–1.41. The CL loss graph demonstrates CL peak from 10.08 dB/cm to 71.12 dB/cm for the range from 1.29 to 1.41. Whereas, the AS graph in Fig. 5

Table 2
Performance metrics of the proposed model for RI-based differentiation of Cancer, dengue, and UTI Samples.

Channel	Category	Specimen	Target	Biological State	RI	Ref.	CL (dB/cm)	WS (nm/RIU)
Ch-1	Cancer	Skin	-	Healthy	1.360	[5]	41.72	-
		Cervical	-	Healthy	1.368	[5]	50.42	-
		Blood	-	Healthy	1.376	[5]	60.31	-
		Adrenal Gland	-	Healthy	1.381	[5]	67.04	-
		Breast	-	Healthy	1.385	[5]	72.46	-
		Breast	-	Healthy	1.387	[5]	74.90	-
	Dengue	Hemoglobin	-	Normal	1.365	[7]	44.54	-
		Platelet	-	Normal	1.390	[7]	38.76	-
		Plasma	-	Normal	1.350	[7]	23.80	-
	UTI	Urine	-	Uninfected	1.335	[6]	22.05	-
Ch-2	Cancer	Skin	Basal	Cancerous	1.380	[5]	66.80	4000
		Cervical	HeLa	Cancerous	1.392	[5]	80.12	6250
		Blood	Jurkat	Cancerous	1.390	[5]	79.25	6428.57
		Adrenal Gland	PC-12	Cancerous	1.395	[5]	80.07	8571.43
		Breast	MDA-MB-231	Cancerous	1.399	[5]	73.10	10000
		Breast	MCF-7	Cancerous	1.401	[5]	66.36	11714
	Dengue	Hemoglobin	-	Infected	1.400	[7]	68.62	7714.29
		Platelet	-	Infected	1.357	[7]	78.12	4848.48
		Plasma	-	Infected	1.337	[7]	34.25	1538.46
	UTI	Urine	Pseudomonas	Infected	1.371	[6]	54.72	2777.78
		Urine	E. coli	Infected	1.388	[6]	76.78	3584.91
		Urine	E. faecalis	Infected	1.3921	[6]	79.78	4028.02

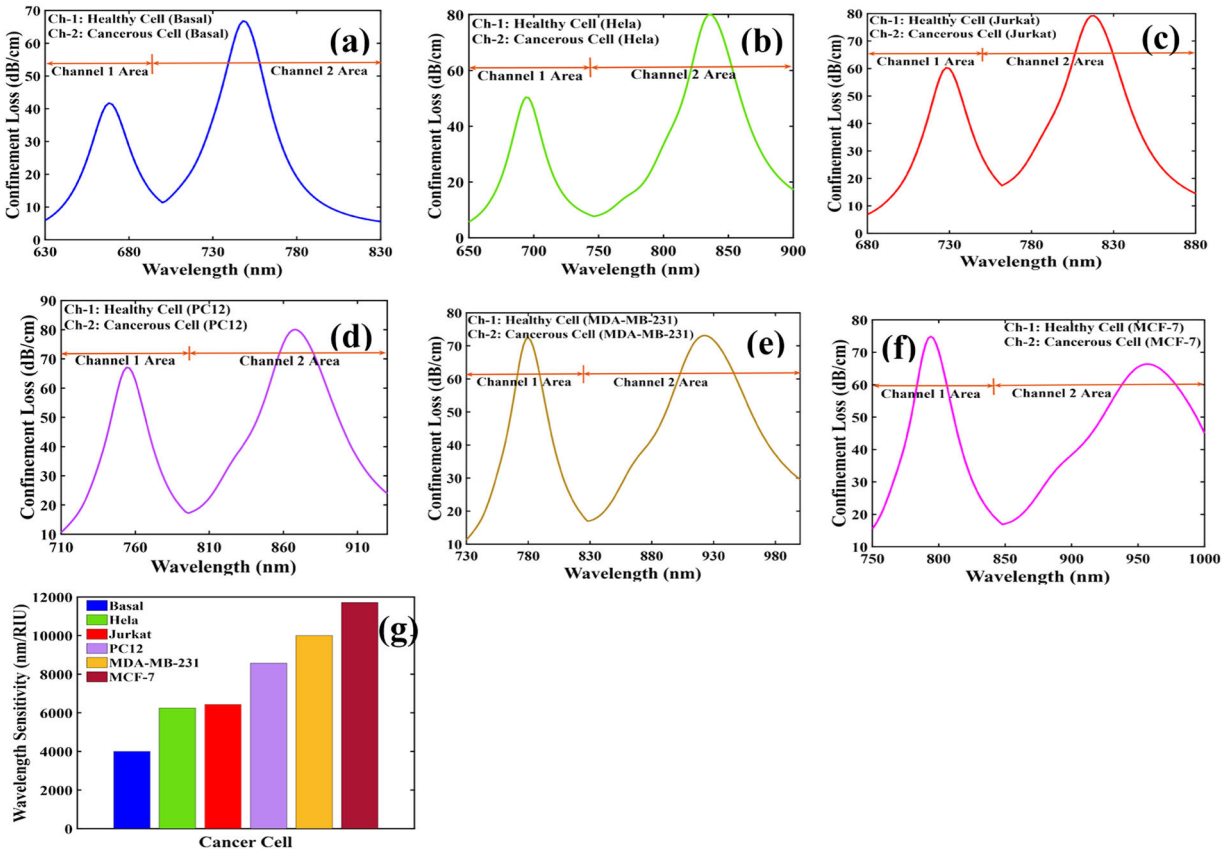


Fig. 8. RI-dependent CL response for differentiating normal biological cells in Ch-1 and cancer-affected (a) Basal, (b) HeLa, (c) Jurkat, (d) PC12, (e) MDA-MB-231, and (f) MCF-7 cells in Ch-2 (g) Bar chart comparison of WS performance for six cancer cell lines.

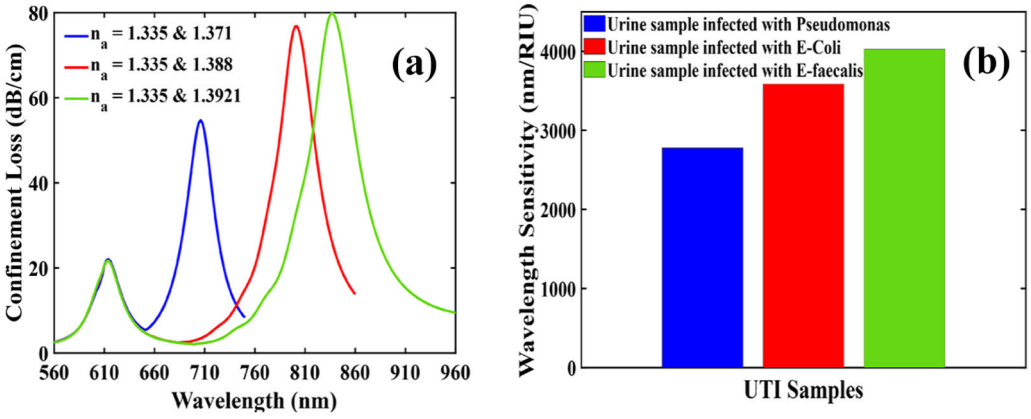


Fig. 9. (a) CL spectra for the normal urine RI in Ch-1 and the RIs of three uropathogen-infected samples in Ch-2; (b) WS performance.

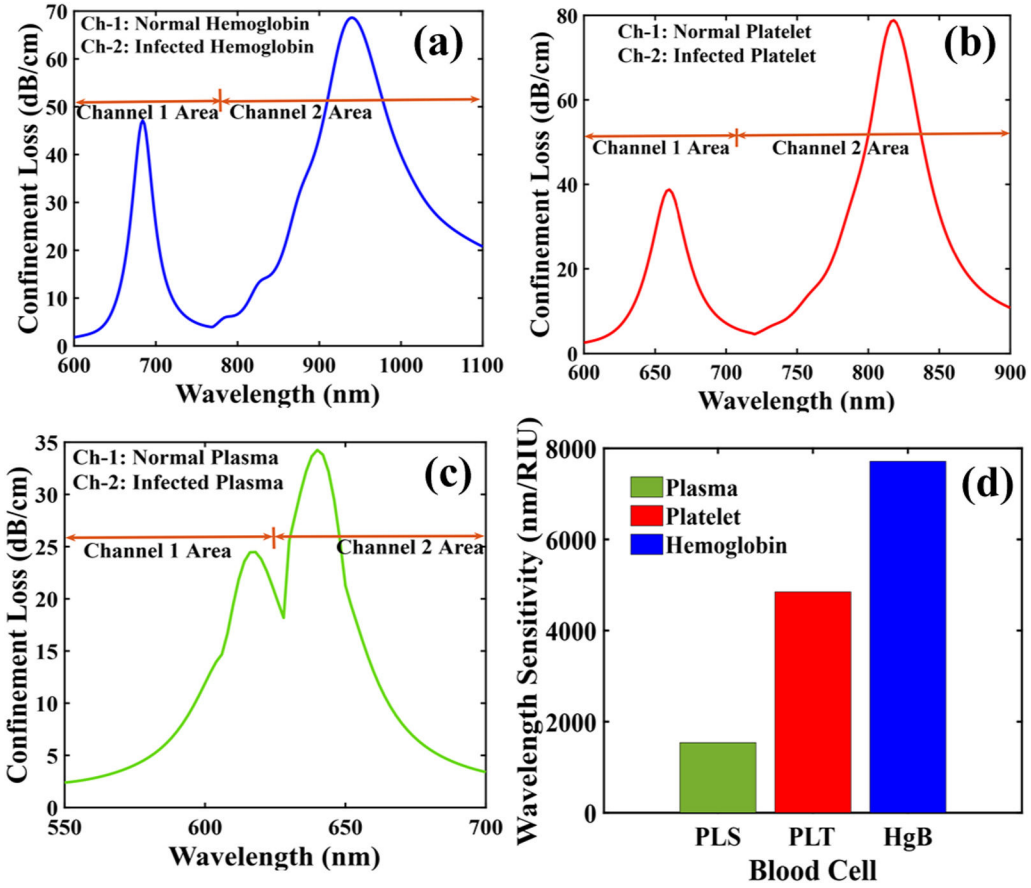


Fig. 10. Simulated CL spectra for the normal blood-component RI in Ch-1 and the dengue-associated (a) hemoglobin, (b) platelet, and (c) plasma RI in Ch-2, and (d) Bar chart comparison of WS performance.

(d) exhibits the maximum AS at 832 RIU^{-1} for RI 1.38 while having a range start from -58 RIU^{-1}. Maximum WS occurs at 1.40 which is 1,22,000 nm/RIU and has a numerical resolution of 8.2×10^{-7} RIU. The complete results showcasing the performance are exhibited in Table 1. As seen from Table 1, the CL increases monotonically with RI up to RI 1.39; after that a decrease can be observed. This indicates an enhanced plasmonic coupling and stronger evanescent field interaction with the sensing medium. The AS and WS show an overall increasing trend with RI.

When evaluating sensor performance, FOM is another key factor to be considered. A higher FOM implies better sensing performance. Fig. 6 shows FOM and FWHM versus RI for the developed sensor in the range 1.17–1.41. The FWHM exhibits a non monotonic

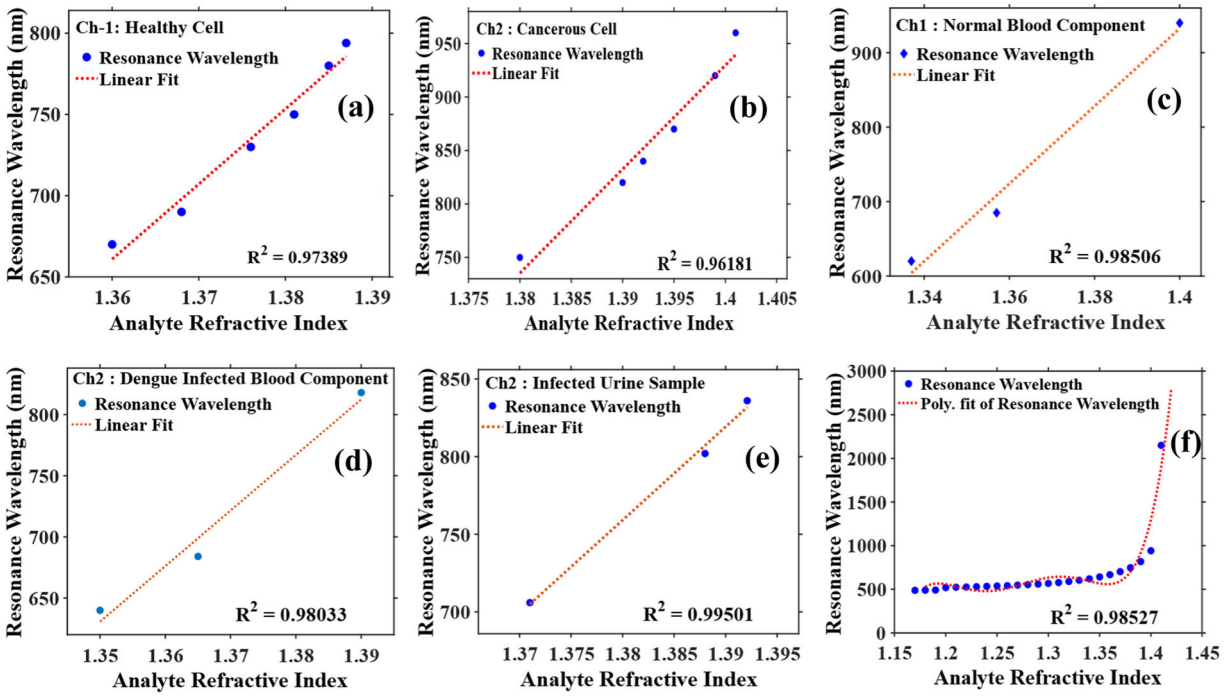


Fig. 11. Linear fitting between the refractive index of different analytes and resonance wavelength for (a) healthy cell, (b) cancerous cell, (c) normal blood component, (d) dengue-infected blood component, (e) infected urine sample, and (f) Polynomial fit curve of different WL for RI range of 1.17–1.41.

dependence on RI. The sensor also maintains a high FOM owing to the substantial enhancement in WS.

4. Simultaneous dual-analyte RI sensing and RI-based discrimination of representative biomedical sample conditions

4.1. Dual channel response of fixed and varied analyte RI

Simultaneous detection of RIs using two RI values in Ch-1 and Ch-2 is shown in Fig. 7. Here, simultaneous refers to two analyte conditions occupying the separate channels concurrently. While detecting simultaneous channels, the left-side channel analyte detects the resonance peak of the lesser RI, while the right side of the channel analyte detects the peak of the larger RI. Fig. 7(a) and 7(c) exhibit the graph for CL loss and AS for Ch-1 fixed and Ch-2 varied. Ch-1 is kept at 1.30 analyte RI while Ch-2 is varied from 1.35, 1.36, 1.37, 1.38, and 1.39. Similarly, Fig. 7(b) and 7(d) exhibit the CL loss and AS graph for Ch-1 varied and Ch-2 fixed. Ch-1 varies from 1.29 to 1.33, while Ch-2 is kept fixed at 1.37.

4.2. RI-Based spectral differentiation of cancer, UTI and dengue associated samples

The proposed fiber does not recognize a particular cell, pathogen or protein; it responds to the effective refractive index of the aqueous or biological medium that fills the channel and overlaps with the evanescent field of the gold-supported SPP mode. The RI values listed in Table 2 are the bulk indices reported in Ref. [5–7] for the six cancer cell lines (basal, HeLa, Jurkat, PC-12, MDA-MB-231 and MCF-7), for the hemoglobin, platelet and plasma fractions of normal and dengue-infected blood, and for uninfected urine and urine infected with Pseudomonas, E. coli and E. faecalis. Figs. 8(a) to 8(f) exhibit the response for different cancer cells. Ch-1 carries the lower reference index of the normal cell, whereas Ch-2 carries the higher index of the corresponding cancerous cell, so that the index contrast appears as a displacement of the confinement-loss peak. Fig. 8(g) exhibits the WS performance between normal cells and malignant cells for all 6 kinds of cancer. The fluidic inlet illustrated in Fig. 1(c) can be used to inject the representative biological samples corresponding to RI values used in this numerical study to the sensing channels. Cell samples for the six cancer-cell types may be obtained from appropriately diagnosed tissue or clinical specimens as follows: Basal cells from skin tissue specimens, HeLa cells from cervical tissue specimens, Jurkat cells can be obtained from blood/hematological specimens, MDA-MB-231 and MCF-7 cells from breast tumor tissue specimens and PC-12 is a rat adrenal pheochromocytoma line included only as an RI test point reported in the literature [5], [8], [9]. Following appropriate sample preparation, the cell-containing biological medium can be delivered to the corresponding sensing channel for RI-based interrogation. Likewise, blood-based components which contain hemoglobin, platelets and plasma from clinically characterized samples may be used for the measurement of dengue-related components [7], and likewise urine samples [6] from clinically characterized infected and uninfected may be used for the measurement of UTI related components. As the

Table 3
Performance comparison with recent studies.

Ref.	RI Range	Simultaneous detection	Max. AS (RIU^{-1})	Max. WS (nm/RIU)	FOM (RIU^{-1})	Resolution	Reporting Year	Application
[15]	1.33–1.40	Yes	-	11600	-	9.71×10^{-6}	2020	Biological proteins, DNA/RNA, and toxic chemicals
[24]	1.29–1.40	No	1189	12500	240	8.0×10^{-6}	2023	-
[25]	1.33–1.43	No	-	123000	683	8.13×10^{-7}	2024	Cancer cell detection
[5]	1.25–1.43	No	1481.1	47,000	350	3.57×10^{-6}	2025	Cancer cell detection
[8]	1.33–1.40	Yes	327.21	7000	-	1.43×10^{-5}	2025	Cancer and Tuberculosis Detection
[12]	1.15–1.41	Yes	1204	10000	-	1×10^{-5}	2026	-
[16]	1.33–1.45	No	2338	32000	1333	3.13×10^{-6}	2026	-
This Study	1.17–1.41	Yes	832.18	122000	739.39	8.2×10^{-7}	2026	Cancer cell, UTI and Dengue detection

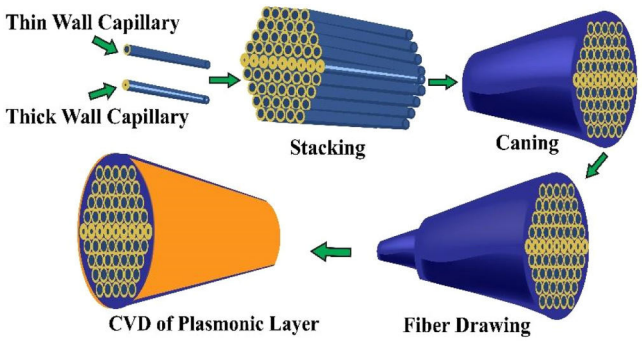


Fig. 12. Illustration of the steps for the conceptual fabrication route of the proposed structure utilizing the stack-and-draw technique.

present work focuses solely on numerical simulation, these methods for sample acquisition and fluidic delivery are indicative and do not correspond to experimental execution of this experiment.

Fig. 9(a) distinguishes normal and 3 urine pathogens via simultaneous detection. Using RI values given in Table 2 and the proposed sensor, we detect the healthy and affected Urinary Tract Infection cells. Using the RI values given in Table 2 and the proposed sensor, we detect Hemoglobin, Platelets, and Plasma of a normal person and a Dengue-affected person. While Ch-1 detects RI with normal Hemoglobin, normal Platelets, and normal Plasma, and Ch-2 detects RI with affected Hemoglobin, affected Platelets, and affected Plasma, as exhibited in Figs. 10(a), 10(b) and 10(c). Whereas, Fig. 10 (d) exhibits the WS performance between normal and dengue-infected blood components.

As observed from the WS performance chart of Fig. 8(g), among all the cancerous cells, the MCF-7 cell exhibited the maximum WS of 11714 nm/RIU. Fig. 9(b) and Fig. 10(d) illustrate that the model achieves 4028.02 nm/RIU for a urine sample infected with E. faecalis and 7714.29 nm/RIU WS for dengue-infected hemoglobin cells, respectively.

The resonance wavelength as illustrated in Figs. 11(a) to 11 (e) exhibits an approximately linear dependence on RI for the healthy-cell, cancerous-cell, normal blood-component, dengue-infected blood-component, and infected-urine datasets, with R^2values of 0.97389, 0.96181, 0.98506, 0.98033, and 0.99501, respectively. The complete simulated RI response over 1.17–1.41 exhibits a nonlinear dependence, for which the polynomial fit gives $R^2 = 0.98527$. These fitting results confirm that the resonance wavelength varies systematically with analyte RI, verifying the suitability of the fitting model.

Table 3 presents a comparison of the proposed sensor with previously reported numerical models. These results indicate a favorable combination of sensing range and peak sensitivity of the proposed design, while the principal advantage of the sensor is its integration of dual-channel operation with broad RI range rather than superiority in every individual performance metric.

5. Proposed fabrication route

The proposed dual-channel PCF–SPR structure is considered at the numerical-design stage, and the following discussion outlines a possible fabrication route following the stack-and-draw technique [5], which is shown in Fig. 12. The thin-walled and thick-walled silica capillaries, along with solid silica rods, are arranged in the hexagonal geometry as shown in Fig. 1(b) to create a preform and the preform is contracted into a cane and further drawn to a fiber shape in a controlled thermal environment. Preservation of the specified submicron air-hole dimensions would depend on the optimization of the preform assembly and drawing conditions. After the formation of the fiber, a thin gold film is considered directly on the outer cladding surface by a vapor-phase technique such as CVD which is used for metal coatings on fibers. The film is applied on the outside cladding and therefore not required to go through the air holes. The conceptual scheme of supplying two analyte media separately to the two corresponding sensing regions is shown in Fig. 1 (c). For one possible implementation, this would necessitate two separate compartments positioned adjacent to their assigned sensing regions, with an impermeable partition or suitable sealing interface preventing lateral communication between compartments. The inlet and outlet manifolds would offer the desired different sample pathways and the fiber–fluidic interfaces can be shaped to keep the fibers aligned and sealed over the active sensing region. The schematic thus forms the proposed fluid-routing idea for the dual-channel optical design. The major fabrication difficulties include collapse of the sub-micron air holes (d_1 and d_c) during drawing due to surface tension, and the spaces between the stacked capillaries. These can be controlled through a regulated draw temperature, gas pressurization of the holes, and vacuum fusion of the gaps. Appropriate sealing at the fiber–microfluidic interface is required to prevent leakage and cross-contamination between the channels.

6. Conclusion

A novel high-performance dual-core, dual-channel PCF-SPR model is developed and numerically evaluated over RI range of 1.17–1.41 in this work. Through optimization of key design parameters, the design achieves a remarkable WS up to 122,000 nm/RIU, a notable AS of 832.18 RIU^{-1}, FOM of 739.39 RIU^{-1}. Using literature-reported RI values ranging between 1.335 and 1.401 for relevant biological states, the study also evaluates the spectral responses associated with six cancer cell lines, three UTI pathogens, and normal

and dengue-associated blood components. The sensor demonstrates a maximum WS of 11714 nm/RIU in the case of MCF-7 cancer cells, 7714.29 nm/RIU for dengue-infected hemoglobin cells, and 4028.02 nm/RIU for a urine sample infected with E. faecalis. Although designed for multi-analyte sensing, the proposed sensor is equally suitable for single-analyte applications. The numerical results show that the proposed dual-channel PCF-SPR structure is a promising platform for wide-range, two-analyte RI sensing. Its application to biomedical samples is so far limited to RI-based discrimination. Future work may focus on experimental fabrication and validation with clinically derived samples to translate the proposed dual-channel PCF-SPR platform from numerical RI-based multi-analyte sensing toward practical biomedical screening.

CRediT authorship contribution statement

Adib Hassan: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Md. Tanjim Hasan:** Writing – review & editing, Software, Investigation. **Shaharior Anik:** Validation, Supervision. **Tanvir Ahmed:** Validation, Supervision. **Paul Alok:** Validation, Supervision.

Declaration of Competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Adib Hassan. His research interests include optical sensing, surface plasmon resonance (SPR) biosensors, photonics, nanomaterials, plasmonics, Sensing and Bio-Sensing Research and advanced sensor design.

Md. Tanjim Hasan. His academic and research activities focus on electrical and electronic engineering, with interests in emerging technologies, advanced sensing systems, and related interdisciplinary research.

Shaharior Anik. His research interests include electrical and electronic engineering, photonics, sensor, intelligent systems, advanced technologies, and interdisciplinary engineering research.

Tanvir Ahmed. His academic interests include electrical engineering, electronics, photonics, Sensing and Bio-Sensing Research

Alok Kumar Paul. His research and academic expertise are related to electrical and electronic engineering and associated technological fields.